

SM-10: First-Principles Quark Mass from Finite Element Chain Network Simulation

600-Cell Standard Model Emergence Series

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Abstract

SM-8 established a zero-free-parameter quark mass formula $M_q = m_e(z/\varphi)V^{7/3}$ that predicts all four heavy quark masses to RMS 2.1%. SM-9 showed that the exponent $7/3$ arises from pair counting $\times$ cage linear dimension, and that the mass is physically distributed across a network of radial and tangential DP chains filling the cage interior. However, the exponent has not been derived from first principles. This paper proposes a finite element method (FEM) simulation to compute quark masses directly: place cage CPs at their lattice-determined positions, fill the interior with DP Sea, let every CP seek opposite-polarity targets to form chains, and sum the total organised DP count $\times M_0$. If the simulation reproduces $V^{7/3}$ scaling without imposing any power law, this would constitute the first derivation of quark mass ratios from geometry alone.

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1 Motivation

The CPP quark mass programme has progressed through three stages:

1. **SM-8 v2.1:** Calibrated formula $m \sim V^{2.38} \times z$ (2 parameters, 0.02% top precision).
2. **SM-8 v4.0:** Zero-parameter formula $M = m_e(z/\varphi)V^{7/3} \times [zC_F]$ (0 parameters, 2.1% RMS).
3. **SM-9 v2.0:** Partial derivation of 7/3 from pair counting; chain-type energy budget decomposition.

The remaining gap: the exponent 7/3 is motivated but not rigorously derived. The chain-type analysis (SM-9 v2.0) showed that the mass arises from a complex volume-filling network of DP chains with three distinct bonding regions, but the exact scaling of this network with cage size has not been computed from first principles.

A direct simulation—placing CPs, forming chains, counting organised DPs—would bypass the need for an analytical derivation and provide a numerical first-principles prediction.

2 Physical Model

2.1 The chain network

A caged quark consists of:

- A central qCP at the cage centre.
- V_{opp} opposite-polarity cage vertices (chain anchors).
- V_{same} same-polarity cage vertices (surface radiators).
- E_{attr} attractive cage edges (tangential chain sites).
- The DP Sea filling the interior volume.

Each CP (whether on a chain or in the Sea) has two charges of opposite polarity. Each charge seeks the nearest opposite-polarity target, forming a chain link. This process cascades: every CP recruited into a chain becomes a new source of lateral connections.

2.2 Three bonding regions

The chain network self-organises into three regions (identified by Abshier):

Region 1 (near central CP): Radial chains converge at small separations. Tangential CPs terminate on the central CP or adjacent radial CPs. Dense cross-linking.

Region 2 (mid-cage): Radials have diverged. Tangential chains extend to neighbouring radials' tangential chains, forming a web mesh. Every DP's two CPs seek opposite-polarity targets in all directions—the “complex melee.”

Region 3 (near cage surface): Tangential chains arch upward toward opposite-polarity cage surface CPs. Organisation converges on cage vertices.

2.3 Surface blanket

Same-polarity cage vertices (V_{same}) launch outward radial chains to the thermalization distance. These surface radials develop their own tangential linking, creating a structure identical to the up quark's radial-plus-tangential blanket. This contributes additional organised DPs.

2.4 Linear oscillator (down-type quarks)

Down-type quarks (d, s, b) include a captured eCP that adds a linear/radial ZBW oscillation mode. This mode contributes mass through its own chain structure.

2.5 Mass calculation

The quark mass equals the total number of organised DPs in the chain network multiplied by the bare DP energy:

$$M_q = M_0 \times N_{\text{organised}} \quad (1)$$

where $M_0 = m_e \times z/\varphi = 3.790$ MeV is the energy per DP link and $N_{\text{organised}}$ is the total count of DPs recruited from the Sea into the chain network.

3 Simulation Design

3.1 Geometry setup

1. Place a central qCP at the origin.
2. Place cage CPs at the lattice-determined positions for the target cage (tetrahedron, icosahedron, dodecahedron, or icosidodecahedron).
3. Assign charge polarity: $V/2$ positive, $V/2$ negative, at the $2/3$ attractive fraction assignment.
4. Fill the interior volume with DP Sea at natural density ρ_{Sea} (to be determined from the lattice spacing l_P or parameterised).

3.2 Chain formation algorithm

At each simulation step:

1. For each CP (central, cage, and Sea), identify the nearest opposite-polarity CP that is not already bonded.

2. Form a chain link (DP bond) between them.
3. Mark both CPs as “organised” (part of the chain network).
4. Newly organised CPs become sources for the next step’s lateral connections.

The simulation terminates when no further chain formation occurs (all nearby opposite-polarity targets are bonded or beyond the thermalization distance).

3.3 Boundary conditions

- **Inner boundary:** The central CP is fixed.
- **Cage boundary:** Cage CPs are fixed at their lattice positions. They act as impedance boundaries (Appendix B of SM-8).
- **Outer boundary:** DPs beyond the thermalization distance r_{therm} are unorganised (remain part of the random Sea). For the top quark, the Shell 3 gap is automatically present in the lattice geometry.

3.4 Output

The primary output is $N_{\text{organised}}$: the total count of DPs recruited into the chain network. Secondary outputs include:

- Spatial distribution of organised DPs (Region 1/2/3 fractions)
- Radial vs. tangential chain ratio
- Surface blanket DP count
- Chain length distribution
- Effective fractal dimension of the network

4 Predictions and Success Criteria

4.1 Primary prediction

If the simulation correctly models the physics, the organised DP counts should satisfy:

$$\frac{N_{\text{organised}}(q_1)}{N_{\text{organised}}(q_2)} = \frac{M_{q_1}}{M_{q_2}} = \frac{V_1^{7/3} \times \text{gap}_1}{V_2^{7/3} \times \text{gap}_2} \quad (2)$$

for all quark pairs. Specifically:

Table 1: Target DP count ratios (normalised to strange).

Ratio	Target	From $V^{7/3}$	From PDG
N_c/N_s	13.0	13.0	13.6
N_b/N_s	42.8	42.8	44.8
N_t/N_s	1762	1762	1850

4.2 Secondary predictions

1. The radial/tangential energy split should be $\sim 44\%/56\%$ across all cages (from SM-9 chain-type analysis).
2. The three-region structure should emerge naturally without being imposed.
3. The effective fractal dimension of the chain network should be between 2 and 3 (corresponding to the $7/3$ exponent).
4. The top quark simulation should show qualitatively different chain structure (long-range coordination bonds through the Shell 3 gap).

4.3 Success criteria

Level A (strong success): DP count ratios match PDG mass ratios to $< 5\%$. The $V^{7/3}$ scaling emerges without being imposed. Fractal dimension measured.

Level B (partial success): DP count ratios show the correct ordering and approximate magnitude. The three-region structure emerges. Scaling exponent is measurable but differs from $7/3$.

Level C (informative failure): The simulation reveals which physical assumptions of the chain model are incorrect, guiding revision of SM-9.

5 Computational Requirements

5.1 Scale estimates

- **Strange quark (tetrahedron):** Confinement sphere radius $\sim 1/\varphi$. Volume ~ 1 (lattice units³). At $\rho_{\text{Sea}} \sim 100$ DPs per unit volume: ~ 100 DPs. Trivial on CPU.
- **Top quark (icosidodecahedron):** Confinement sphere radius $\sim \sqrt{2}$. Volume ~ 12 (lattice units³). At same density: ~ 1200 DPs. Plus surface blanket: ~ 2000 total. Still CPU-tractable.
- **With fine-grained DP Sea ($\rho \sim 10^4$):** $\sim 10^4$ – 10^5 DPs. GPU recommended.

5.2 Implementation options

Phase 1 (CPU, proof of concept): Coarse-grained simulation with $\rho \sim 100$. Test whether DP count ratios track $V^{7/3}$. Python with NumPy. Estimated time: 1–2 sessions.

Phase 2 (GPU, production): Fine-grained simulation with $\rho \sim 10^4$. Measure fractal dimension, region fractions, chain statistics. CUDA or JAX. Estimated time: 1–2 weeks.

Phase 3 (lattice-scale): Full 600-cell tessellation with explicit lattice CPs. Test whether the simulation reproduces the Shell 3 gap effect and the $z \times C_F = 16$ multiplier. GPU cluster. Estimated time: months.

6 Relationship to Existing Methods

Lattice QCD: Computes quark masses on a discretised spacetime lattice using Monte Carlo sampling of the path integral. Requires the QCD Lagrangian as input. CPP’s FEM simulation uses a different dynamical rule (CP charge-seeking rather than gauge field evolution) and a different lattice (600-cell rather than hypercubic).

Bag models: Confine quarks within a spherical boundary with phenomenological boundary conditions. The CPP cage plays a similar role but the boundary is a polyhedral lattice substructure, not imposed.

String models: Model confinement via flux tubes between quarks. CPP’s radial chains are the microscopic realisation of the flux tube, with the chain composed of explicit DP links.

The CPP FEM simulation is distinguished by:

1. The lattice is derived from geometry (600-cell), not chosen for computational convenience.
2. The dynamical rule (charge-seeking) is simpler than gauge field evolution.
3. The mass prediction is a DP count, not an eigenvalue of a transfer matrix.
4. Zero free parameters in the mass formula (if the simulation reproduces $V^{7/3}$).

7 Open Questions for the Simulation

1. **What is ρ_{Sea} ?** The DP Sea density should be derivable from the 600-cell lattice spacing and the ZBW frequency. Is it $1/t_{\text{edge}}^3$? z/t_{edge}^3 ? This is the only parameter not yet fixed.
2. **Does the gap multiplier emerge?** For the top quark, the Shell 3 gap should force chains to use coordination bonds. Does the simulation naturally produce a $\times 16$ enhancement, or does it require explicit modelling of the colour factor C_F ?
3. **What sets the thermalization distance?** Chains terminate when the organised DP’s SSV field falls below the Sea’s thermal noise. This distance should be computable from $\alpha_{\text{geom}} = 1/\sqrt{5}$.
4. **Does the exponent drift?** SM-9 documented a 2% drift in the pairwise exponent across pre-gap quarks. Does the simulation reproduce this drift, and does it reveal the physical mechanism?
5. **Can the simulation distinguish 7/3 from $3 - 1/\varphi$?** Grok’s conjectured exponent (2.382) differs from 7/3 (2.333) by 2%. A simulation with sufficient precision could distinguish them.

8 Timeline and Milestones

Table 2: Proposed milestones for SM-10.

Phase	Milestone	Deliverable	Est. time
0	Simulation design document	This paper (v0.1)	Done
1a	CPU proof-of-concept	DP counts for all 4 cages	1–2 sessions
1b	Ratio test	Check N_q ratios vs $V^{7/3}$	Same session
2a	GPU implementation	CUDA/JAX chain-formation kernel	1 week
2b	Production runs	4 cages $\times$ 10 densities	1 week
2c	Analysis	Fractal dimension, region fractions	1 session
3	Paper draft	SM-10 v1.0	1–2 sessions

9 Conclusion

The FEM chain network simulation is the natural next step for the CPP quark mass programme. It would convert the scaling formula $M = m_e(z/\varphi)V^{7/3}$ from a calibrated observation into a first-principles computation, and it would test Thomas Abshier’s physical picture of three-region bonding, fractal cascading, and surface blanket formation. If successful, it would be the first derivation of quark mass ratios from geometry in any theoretical framework.

The Phase 1 proof-of-concept (CPU, coarse-grained) can be attempted immediately. If ratios are promising, Phase 2 (GPU, fine-grained) would follow.

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The CPP programme is registered at OSF (DOI: <https://doi.org/10.17605/OSF.IO/JXE8D>) and maintained at GitHub (<https://github.com/Hyperphysics-Institute/Cpp>).

References